PS-05: Personal Finance Agent

Domain: FinTech

Build an AI financial assistant that analyzes bank statements, categorizes spending, detects anomalies, tracks budgets, and provides proactive financial insights and alerts.

In essence:
An always-on personal finance advisor that monitors spending and improves financial health.

Build an Autonomous Financial Wellness Agent

Develop a multi-agent AI system capable of:

- Understanding financial transactions.
- Categorizing expenses automatically.
- Tracking spending patterns.
- Detecting unusual activity.
- Monitoring budgets.
- Forecasting future cash flow.
- Generating personalized financial advice.
- Alerting users before financial problems occur.

The system should function as a continuous financial advisor rather than a passive budgeting application.

Example Scenario

User Uploads

3 Months Bank Statements

or

CSV

PDF
UPI Transaction History
Credit Card Statements

Traditional Finance App

Food: ₹12,000

Travel: ₹3,000

Shopping: ₹6,000

Agentic Finance System

Detected:

Food expenses increasing 18% monthly.

3 unused subscriptions found.

Electricity bill unusually high.

Projected savings this month:
₹6,500

Potential annual savings:
₹22,000

The system actively improves financial health.

Core Components

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1. Financial Data Ingestion Agent

First responsibility:

Collect and normalize financial data.

Sources:

Bank Statements
Credit Cards
UPI Transactions
Wallets
Investment Accounts

Example:

HDFC Statement.pdf

Converted into:


```
{
 "date":"2026-02-15",
 "amount":250,
 "merchant":"Swiggy"
}
```


All data becomes structured.

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2. Transaction Categorization Agent

Automatically classifies spending.

Example:

Swiggy

↓

Food & Dining

Uber

↓

Transportation

Netflix

↓

Entertainment

Possible categories:

Food
Travel
Healthcare
Shopping
Bills
Education
Investments
Subscriptions

The system continuously improves classification accuracy.

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3. Spending Pattern Analysis Agent

This is where intelligence begins.

Example:

Food Spending

January: ₹5,000
February: ₹6,500
March: ₹8,000

Agent detects:

Consistent increase trend.

Another example:

Fuel Spending

Usually: ₹2,000

Current Month: ₹6,500

Flag anomaly.

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4. Budget Management Agent

Creates and manages budgets.

Example:

Monthly Income

₹50,000

Recommended:

Needs: ₹25,000

Wants: ₹15,000

Savings: ₹10,000

The budget evolves with spending behavior.

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5. Financial Health Scoring Agent

Think of this as:

Credit Score
+
Budget Discipline
+
Savings Consistency

Combined into:

Financial Health Score

82 / 100

Factors:

- Savings rate
- Debt ratio
- Spending consistency
- Emergency fund coverage

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6. Anomaly Detection Agent

One of the strongest features.

Example:

Normal Spending:
₹500/day

Suddenly:

₹15,000 transaction

Agent flags:

Potential Fraud

Another example:

Netflix billed twice

↓

Duplicate Charge Detected

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7. Subscription Intelligence Agent

Most users forget subscriptions.

Detect:

Netflix
Spotify
Amazon Prime
ChatGPT

Identify:

Unused subscriptions

Example:

Spotify subscription active

No transactions or usage indicators for 4 months

Recommendation:
Cancel

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8. Cash Flow Forecasting Agent

Very valuable for demos.

Based on:

Income
Bills
EMIs
Recurring Expenses

Predict:

Expected Balance

Example:

Current Balance:
₹32,000

Predicted Balance after 25 days:
₹4,500

Alert user before financial stress occurs.

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9. Financial Advisor Agent

This is the most "agentic" component.

Instead of reporting:

Spent ₹10,000 on food

Recommend:

Reducing restaurant visits by 2 per week could save approximately ₹18,000 annually.

Another example:

Emergency fund covers only 1.2 months.

Recommended:
Build reserve for 6 months.

The system becomes a financial coach.

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10. Alert & Notification Agent

Provides proactive alerts.

Examples:

Budget Risk

⚠ Food budget 90% utilized.

Fraud Alert

⚠ Unusual spending detected.

Subscription Alert

⚠ Unused subscription detected.

Savings Alert

⚠ Savings rate dropped below target.

Multi-Agent Architecture

A serious implementation might include:

Agent 1

Data Ingestion Agent

Agent 2

Transaction Categorization Agent

Agent 3

Spending Analysis Agent

Agent 4

Budget Planning Agent

Agent 5

Financial Health Agent

Agent 6

Anomaly Detection Agent

Agent 7

Subscription Intelligence Agent

Agent 8

Cash Flow Forecast Agent

Agent 9

Financial Advisor Agent

Agent 10

Notification Agent

Agent 11

Master Orchestrator

Coordinates the entire workflow.

Advanced Features (What Can Win)

1. Goal-Based Planning

User sets:

Buy Car in 2 Years

Agent calculates:

Need to save ₹8,500/month.

Tracks progress automatically.

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2. Life Event Simulation

Example:

What if I buy a ₹12 lakh car?

Agent simulates:

EMI Impact
Savings Impact
Cash Flow Impact

Very impressive in demos.

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3. Financial Twin

Create a digital representation of the user's finances.

The agent can test:

New House
New Loan
Salary Increase
Job Loss

and predict outcomes.

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4. Investment Readiness Agent

Instead of recommending specific investments, assess readiness.

Example:

Emergency Fund:
Incomplete

Recommendation:
Complete emergency savings before increasing investment risk.

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5. Family Finance Dashboard

Track:

Parents
Spouse
Children

Combined financial health.

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6. AI CFO Dashboard

Display:

Financial Health Score

Monthly Savings

Budget Utilization

Fraud Alerts

Cash Flow Forecast

Upcoming Bills

Excellent judging demo.

What Judges Will Likely Reward

Not:

Statement
↓
Categorization
↓
Charts

But:

Financial Data
↓
Understand Spending
↓
Detect Patterns
↓
Predict Risks
↓
Recommend Actions
↓
Monitor Continuously

↓

Improve Financial Health